

Boundary Control of 2×2 weakly hyperbolic systems: application to wastewater treatment

Presented by: MOHAND OUIDIR AMIRAT

Under Supervision of:

CLAIRE VALENTIN

VINCENT ANDRIEU

MATHIEU BAJODEK

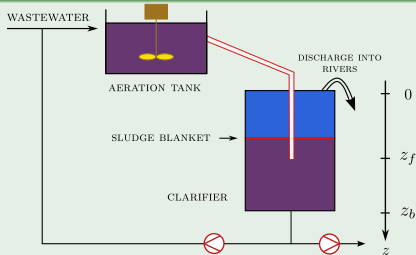
LAGEP



EEA
ÉLECTRONIQUE
ÉLECTROTECHNIQUE
ET AUTOMATIQUE
UNIVERSITÉ DE LYON



Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f} \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \underbrace{g \left(1 - \frac{\rho_l}{\rho_s} \right) \varepsilon_s}_{\text{Apparent weight}} + \underbrace{\frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)}}_{\text{Friction}} + \underbrace{\frac{f_{2s}}{\rho_s} \delta_{z_f}}_{\text{Feed}} \end{cases}$$

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0, t) = U_0(t), \quad f_s(z_b, t) = U_1(t)$$

U_0, U_1 are *signal commands*.

→ Coupling the equations with (**BCs**) corresponds to a *Boundary Control System*

Goal 1: Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth

Goal 2: Prove the stability of the controlled system

OUTLINE OF TALK

- 1 SYSTEM ANALYSIS
- 2 THROUGH A SIMPLIFIED CLARIFIER
- 3 NUMERICAL RESULTS
- 4 CONCLUSIONS
- 5 PERSPECTIVES: STABILIZATION OF A SHOCK STEADY-STATE

SYSTEM ANALYSIS

SYSTEM ANALYSIS

Governing equations

$\mathcal{D} = [0, z_b] \times \mathbb{R}_+ /$ Find $\varepsilon_s, f_s: \mathcal{D} \rightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f}, \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = g \left(1 - \frac{\rho_l}{\rho_s} \right) \varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \frac{f_{2s}}{\rho_s} \delta_{z_f}, \\ f_s(0, t) = U_0(t), \quad f_s(z_b, t) = U_1(t), \\ \varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z, 0) = f_s^0. \end{cases} \quad (1)$$

with:

$$\sigma_e(\varepsilon_s) = \begin{cases} 0, & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}, & \varepsilon_s > \varepsilon_c \end{cases}$$

ρ_l : water density ($:= 1000 \frac{\text{kg}}{\text{m}^3}$)

g : gravity acceleration constant ($:= 9.81 \frac{\text{m}}{\text{s}^2}$)

$\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s$ (parameters)

SYSTEM ANALYSIS

$$\underbrace{\partial_t \begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \underbrace{\partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right)}_{\mathbf{F}(\mathbf{Y})} = \underbrace{\begin{pmatrix} 0 \\ g(1 - \frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \frac{f_{2s}}{\rho_s} \delta_{zf} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A}(\mathbf{Y}) \cdot \partial_z \mathbf{Y} = \mathbf{S}(\mathbf{Y}) \quad (2)$$

where

$$\mathbf{A}(\mathbf{Y}) := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A}(\mathbf{Y})$?

$$\text{Sp}(\mathbf{A}) = \left\{ v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}}, v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}} \right\}, \text{ if } \varepsilon_s > \varepsilon_c$$

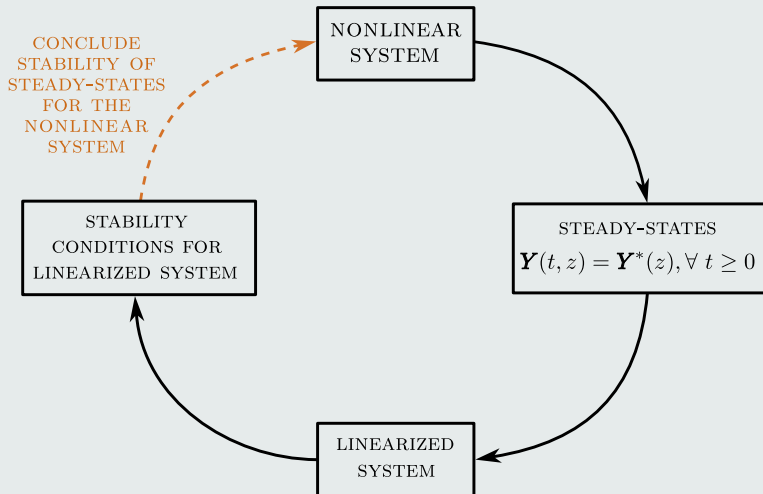
⇒ Strictly hyperbolic system

$$\text{Sp}(\mathbf{A}) = \{v_s, v_s\}, \text{ if } \varepsilon_s \leq \varepsilon_c$$

⇒ Weakly hyperbolic system

SYSTEM ANALYSIS

How to stabilize ? (Indirect Lyapunov's Method)



NONUNIFORM SHOCK STEADY-STATE

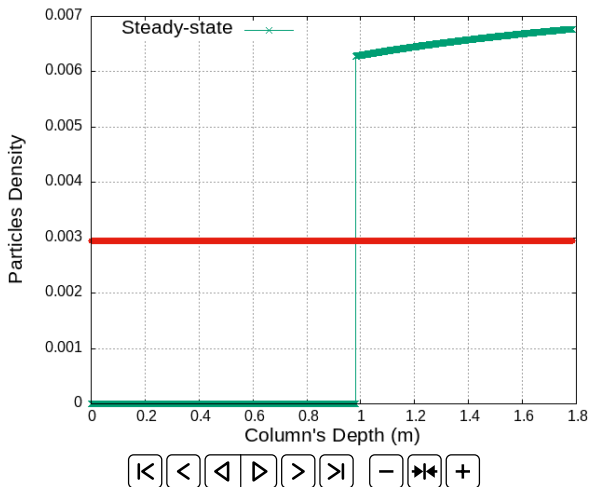


Fig. Simulation of the particles density on 6400 cells using Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state

SYSTEM ANALYSIS: LINEARIZATION

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t, x) - \varepsilon_s^*(x), \quad V_s := v_s(t, x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \quad (3)$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma'_e(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (3) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

Not uniformly possible (due to regions of weak-hyperbolicity)

Rather transform $\mathbf{A}^*(z)$ into a *Jordan* block:

$$\mathbf{A}^*(z) \xleftrightarrow{\psi} \begin{pmatrix} \mathbf{c}_1 & \mathbf{c}_2 \\ \mathbf{0} & \mathbf{c}_3 \end{pmatrix}$$

THROUGH A SIMPLIFIED CLARIFIER

The case of constant coefficients without source term

$$\begin{cases} \partial_t \phi - \begin{pmatrix} \lambda & c \\ 0 & \lambda \end{pmatrix} \partial_x \phi = 0, & t \in (0, +\infty), x \in (0, 1), \\ \phi(0, x) = \phi^0(x), & x \in (0, 1), \end{cases} \quad (4)$$

where $\phi = (\phi_1, \phi_2)^\top : [0, +\infty) \times [0, 1] \rightarrow \mathbb{R}^2$ is the state.

— $\lambda < 0$ is the **transport velocity**, $c \neq 0$ is the **coupling coefficient**.

GOAL

Characterize and identify dissipative boundary conditions which guarantee bounded smooth solutions converging to steady-state $\phi \equiv 0$?

First, assume that system (4) is endowed with the following *linear static boundary conditions*:

$$\phi(t, 1) = \begin{pmatrix} 0 & k_1 \\ k_2 & 0 \end{pmatrix} \phi(t, 0), \quad (5)$$

where k_1, k_2 are constant real coefficients.

Definition 1 (Hale and Lunel)

The steady-state of a system is said to be *stable* if and only if its poles are located in the half plane $(-\infty, 0) \times \mathbb{R}$. Otherwise, the steady-state is said to be *unstable*.

We emphasize a fundamental property of this class of systems, stated in the following theorem.

THEOREM 1

If $c \neq 0$ and $k_2 \neq 0$, then the origin of system (4),(5) is unstable in the sense of Definition 1.

Sketch of the proof

Taking the Laplace transform of system (4)-(5) we obtain the following representation in the Laplace domain (with $s \in \mathbb{C}$):

$$\begin{cases} \frac{d\bar{\phi}}{dx}(s, x) = \begin{pmatrix} \lambda & c \\ 0 & \lambda \end{pmatrix}^{-1} s \bar{\phi}(s, x), \\ \bar{\phi}(s, 1) = \begin{pmatrix} 0 & k_1 \\ k_2 & 0 \end{pmatrix} \bar{\phi}(s, 0), \end{cases} \tag{6}$$

where $\bar{\phi}(s, x) = \mathcal{L}(\phi(\cdot, x))(s)$.

Solving system (6) yields the following *characteristic equation* of the poles of the system

$$p_1(z, e^z) \triangleq e^{2z} - \frac{ck_2}{\lambda} ze^z - k_1 k_2 = 0. \quad (7)$$

p_1 is a quasipolynomial in two variables.

First, what do we know about quasipolynomials?

L. Pontryagin gave necessary and sufficient stability conditions for quasipolynomials in two variables with real or complex coefficients of the following form

$$p(z, e^z) \triangleq \sum_{h=0}^m \sum_{k=0}^n a_{hk} z^h e^{kz}, \quad \text{s.t.} \quad (8)$$

(1) there exists $k \in \{0, \dots, n\}$ such that $a_{mk} \neq 0$,

(2) there exists $h \in \{0, \dots, m\}$ such that $a_{hn} \neq 0$.

The quasipolynomial $p(z, e^z)$ is said to have a principal term if $a_{mn} \neq 0$.

Stability result (due to Pontryagin): If the quasipolynomial $p(z, e^z)$ as in (8) does not have a principal term, then the equation $p(z, e^z) = 0$ has an **infinite number of roots with arbitrarily large positive real parts**.

Thus in this context, our quasipolynomial for the zeros for the system can be expressed as

$$p_1(z, e^z) = e^{2z} - \frac{ck_2}{\lambda} ze^z - k_1 k_2 = \sum_{h=0}^1 \sum_{k=0}^2 a_{hk} z^h e^{kz} \quad (9)$$

Identifying the nonzero coefficients a_{hk} we get

$$a_{00} = -k_1 k_2, \quad a_{02} = 1, \quad a_{11} = -\frac{ck_2}{\lambda} \quad (10)$$

⇒ The quasipolynomial p_1 does not have a principal term (coefficient $a_{12} = 0$),
 ⇒ $p_1(z, e^z)$ admits infinitely many roots $z \in \mathbb{C}$ with *arbitrarily large positive real parts*, and thus system (4), (5) is **unstable**.

Remarks

If we suppose $c = 0$ (corresponding to the *strictly hyperbolic* case) then

$$p_1(z) = e^{2z} - k_1 k_2 = 0, \text{ s.t. } \operatorname{Re}(z) < -0 \Leftrightarrow |k_1 k_2| < 1.$$

As we address the *weakly hyperbolic* case (i.e. for $c \neq 0$), the only way to remain stable under linear static boundary conditions is by enforcing $k_2 = 0$.

However, in a practical engineering situation (for instance the clarifier settling in the context of Wastewater Treatment), imposing $k_2 = 0$ is very restrictive!

Relaxing the constraint $k_2 = 0$?

⇒ Design a wider class of boundary conditions e.g. of *mixed (static/dynamic) type*

Replace,

$$\phi(t, 1) = \begin{pmatrix} 0 & k_1 \\ k_2 & 0 \end{pmatrix} \phi(t, 0), \quad (11)$$

By

$$\phi_1(t, 1) = k_1 \phi_2(t, 0), \quad \frac{d\phi_2}{dt}(t, 1) = k_2 \phi_1(t, 0) - e_p \phi_2(t, 1), \quad (12)$$

where $e_p > 0$ is a constant damping coefficient at the boundary $x = 1$.

Indeed, taking Laplace transform of system (4) along with the mixed boundary condition (12), yields the following characteristic equation of the poles:

$$p_2(z, c) \triangleq (\lambda z + e_p) e^{2z} - \frac{ck_2}{\lambda} z e^z - k_1 k_2 = 0. \quad (13)$$

Remark

Now, the principal term is nonzero, suggesting that the system (4), (12) can be stable for certain nonzero values of c and k_2 , as clearly stated in the following theorem.

THEOREM 2

Let $k_2 \neq 0$, and the damping coefficient e_p satisfy $e_p > |k_1 k_2|$.

We define a critical value $c^* > 0$, depending on the parameters λ, k_1, k_2, e_p , given by

$$c^* := \frac{\lambda}{|k_2|} \min_{z \in D} \frac{|(\lambda z + e_p) e^{2z} - k_1 k_2|}{|z e^z|}, \quad (14)$$

where $D = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0, |z| \leq R_0\}$ with R_0 sufficiently large.

Let $c_{\text{crit,spec}} = \min(c^*, \frac{\lambda^2}{|k_2|})$, Then, for all nonzero values $c \in \mathbb{R}^*$ such that $|c| < c_{\text{crit,spec}}$, the origin of system (4), (12) is stable in the sense of Definition 1.

Sketch of the proof

We rewrite the quasipolynomial (13) in the following way:

$$p_2(z, c) = f(z) + g(z, c), \quad \text{where}$$

$$f(z) = (\lambda z + e_p) e^{2z} - k_1 k_2, \quad g(z, c) = -\frac{ck_2}{\lambda} z e^z.$$

Moreover, let us fix $R > 0$ and define the contour

$$S_R = \{Re^{i\theta} : -\pi/2 \leq \theta \leq \pi/2\}.$$

As $R \rightarrow \infty$, S_R tends to the boundary of the closed right half-plane.

Step 1: We first show that $f(z)$ has no zeros in the closed right half-plane $\overline{\mathbb{C}_+}$,

Step 2: Establish a uniform bound on $\frac{|g(z, c)|}{|f(z)|}$,

Step 3: Apply Rouché's Theorem for counting zeros. ■

We now examine how the stability analysis can be extended to the case where the characteristic velocities λ_i and the coupling coefficient c are functions of the space coordinate x , and in presence of source term.

General case of balance laws with nonuniform coefficients

$$\begin{cases} \partial_t \phi - \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix} \partial_x \phi = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix} \phi, & t \in [0, +\infty), x \in (0, 1), \\ \phi_1(t, 1) = k_1 \phi_2(t, 0), \quad \frac{d\phi_2}{dt}(t, 1) = k_2 \phi_1(t, 0) - e_p \phi_2(t, 1), \\ \phi(0, x) = \phi^0(x), \quad x \in (0, 1). \end{cases} \quad (15)$$

$c(x) \neq 0$, $\forall x \in [0, 1]$, and $\lambda_{1,2}(x) > 0$ s.t. $\exists x \in [0, 1]$ for which $\lambda_1(x) = \lambda_2(x)$

Sufficient conditions ensuring exponential stability of system (15) ?

State space: $X = \mathbb{L}^2((0, 1); \mathbb{R}) \times \mathbb{H}^1((0, 1); \mathbb{R}) \times \mathbb{R}$.

Definition

The origin of system (15) is globally exponentially stable for the X -norm if there exist $\delta > 0$ and $L > 0$ such that, for every $\phi^0 \in X$, the solution of the Cauchy problem satisfies

$$\|\phi(t, \cdot)\|_X \leq L e^{-\delta t} \|\phi_0\|_X, \quad \forall t \in [0, +\infty). \quad (16)$$

Assumption 1

There exists a function $\mathfrak{b} \in C^1([0, 1], \mathbb{R}^*)$ such that $B_{21}(x) = \mathfrak{b}(x)(\lambda_2(x) - \lambda_1(x))$, $\forall x \in [0, 1]$.

Assumption 2

There exists a positive definite map $Q \in C^1([0, 1]; S_2(\mathbb{R}))$, functions $g_1 \in C^1([0, 1]; \mathbb{R}_+^*)$, $g_2 \in C^1([0, 1]; \mathbb{R})$ and a real constant $\beta > 0$ such that the following holds:

i) The map Q has the following form

$$Q(x) = \begin{pmatrix} q_1(x) & q_3(x) \\ q_3(x) & q_2(x) \end{pmatrix}. \quad (17)$$

and satisfies

$$\text{a) } q_1(x) > \frac{g_2^2(x)}{g_1(x)}, \quad \forall x \in [0, 1]. \quad (18)$$

$$\text{b) } q_1(x)q_2(x) > q_3^2(x) + \frac{g_2^2(x)}{g_1(x)}q_2(x), \quad \forall x \in [0, 1]. \quad (19)$$

ii) For all $x \in (0, 1)$, the matrix $M_1(x) \in \mathcal{M}_{3 \times 3}(\mathbb{R})$ is negative definite:

$$M_1(x) = \begin{pmatrix} T(x) & v(x) \\ v^\top(x) & e(x) \end{pmatrix} \prec 0, \quad (20)$$

where its components are given by

$$T(x) = -\text{He}(Q\Lambda)_x + 2\text{He}(QB) - (F_0)_x + F_1, \quad (21)$$

$$v(x) = \begin{pmatrix} q_1 c + g_1 B'_{21} + g_2 \text{tr}(B) - g'_2 \lambda_2 \\ 0 \end{pmatrix}, \quad (22)$$

$$e(x) = g_1(\lambda'_2 + 2B_{22}) - g'_1 \lambda_2 + 2g_2 c. \quad (23)$$

The auxiliary matrices $F_0(x)$ and $F_1(x)$ are defined as

$$F_0 = \begin{pmatrix} 0 & 0 \\ 0 & q_3 c \end{pmatrix}, \quad (24)$$

$$F_1 = \begin{pmatrix} g_2 B'_{21} - g'_2 B_{21} & g_2 B'_{22} \\ \star & -(g_1 B'_{22})_x - (g_2 B_{12})_x \end{pmatrix}, \quad (25)$$

iii) The boundary coupling matrix $M_2 \in \mathcal{M}_{3 \times 3}(\mathbb{R})$ is negative definite:

$$M_2 = \begin{pmatrix} \mathbf{F}_3 & \mathbf{F}_4 \\ \mathbf{F}_4^\top & \mathbf{F}_5 \end{pmatrix} \prec 0, \quad (26)$$

We have that $\mathbf{F}_3 = \mathbf{F}_{3a} + \mathbf{F}_{3b}$, with:

$$\mathbf{F}_{3a} = \begin{pmatrix} -q_1(0)\lambda_1(0) + \frac{g_1(1)}{\lambda_2(1)}k_2^2 & -q_3(0)(\lambda_1(0) + \lambda_2(0)) \\ * & -q_2(0)\lambda_2(0) - q_3(0)c(0) \\ & + k_1^2 q_1(1)\lambda_1(1) \end{pmatrix}, \quad \mathbf{F}_{3b} = \begin{pmatrix} -g_2(0)B_{21}(0) & & \\ + \frac{g_2^2(0)}{g_1(0)}\lambda_2(0) & g_2(1)k_1k_2 - \frac{g_1(1)}{\lambda_2(1)}k_1k_2B_{21}(1) & \\ & g_2(1)B_{21}(1)k_1^2 - g_2(0)B_{12}(0) & \\ * & + \frac{g_1(1)}{\lambda_2(1)}k_1^2B_{21}^2(1) - g_1(0)B_{22}'(0) & \\ & -2g_2(1)B_{21}(1)k_1^2 & \end{pmatrix}.$$

The remaining blocks are:

$$\mathbf{F}_4 = \begin{pmatrix} \frac{\beta k_2}{2} - k_2(e_p + B_{22}(1))\frac{g_1(1)}{\lambda_2(1)} \\ k_1q_3(1)(\lambda_1(1) + \lambda_2(1)) \\ -k_1(e_p + B_{22}(1))\left(g_2(1) - \frac{g_1(1)}{\lambda_2(1)}B_{21}(1)\right) \end{pmatrix},$$

$$\begin{aligned} \mathbf{F}_5 = & -\beta e_p + q_2(1)\lambda_2(1) + q_3(1)c(1) \\ & + g_1(1)B_{22}'(1) + \frac{(e_p + B_{22}(1))^2}{\lambda_2(1)}g_1(1) \\ & + g_2(1)B_{12}(1). \end{aligned} \quad (27)$$

Theorem 3

If Assumptions 1 and 2 holds, then the origin of system (15) is globally exponentially stable.

Sketch of the proof:

We consider the candidate Lyapunov functional $\mathcal{V} = \sum_{i=1}^4 \mathcal{V}_i$ where

$$\mathcal{V}_1 = \int_0^1 \phi^\top Q(x) \phi \, dx,$$

$$\mathcal{V}_2 = \int_0^1 g_1(x) (\partial_x \phi_2)^2 \, dx,$$

$$\mathcal{V}_3 = 2 \int_0^1 g_2(x) \phi_1 \partial_x \phi_2 \, dx,$$

$$\mathcal{V}_4 = \frac{1}{2} \beta \phi_2(t, 1)^2.$$

Let us define $w_1 = [\phi_1, \phi_2, \partial_x \phi_2]^\top$, and $w_2(t) = [\phi_1(t, 0), \phi_2(t, 0), \phi_2(t, 1)]^\top$. Time derivative of the the Lyapunov functional along classical solutions yields

$$\begin{aligned} \dot{\mathcal{V}} &\leq \int_0^1 w_1(x)^\top M_1(x) w_1(x) \, dx + w_2(t)^\top M_2 w_2(t) \\ &\quad + \underbrace{2 \int_0^1 \left(g_1(x) B_{21}(x) - g_2(x) (\lambda_2(x) - \lambda_1(x)) \right) \partial_x \phi_1 \partial_x \phi_2 \, dx}_{\text{Cross-term to cancel}}. \end{aligned}$$

By virtue of Assumption 1, we have $B_{21}(x) = b(x)(\lambda_2(x) - \lambda_1(x))$. to make the term vanish, we select the weight function $g_2(x)$ such that:

$$g_2(x) = b(x)g_1(x), \quad \forall x \in [0, 1].$$

This choice is central to the proof, as it eliminates the cross-term $\partial_x \phi_1 \partial_x \phi_2$. According to Assumption 2, the matrices $M_1(x)$ and M_2 are negative definite. This implies that there exist constants $\delta_1, \delta_2 > 0$ such that:

$$\dot{V} \leq -\delta_1 \int_0^1 (|\phi_1|^2 + |\phi_2|^2 + |\partial_x \phi_2|^2) dx - \delta_2 |\phi_2(t, 1)|^2.$$

By Use of Coercivity + Grönwall Lemma and equivalence of norms we get

$$V(t) = \mathcal{V}(\phi(t; \phi_0)) \leq e^{-\delta t} \mathcal{V}(\phi^0), \quad \forall \phi^0 \in X.$$



Compare the Lyapunov approach with the spectral approach

Consider special case of system (15) with constant coefficients ($\lambda_1(x) \equiv \lambda_2(x) \equiv \lambda > 0$, $c(x) \equiv c \neq 0$) and no source terms ($B(x) \equiv 0$).

Specifically, we choose weight functions of the form:

$$\begin{aligned} q_1(x) &= \alpha_1 e^{\mu x}, & q_2(x) &= \alpha_2 e^{\mu x}, \\ q_3(x) &= 0, & g_1(x) &= \alpha_g e^{\mu x}, \end{aligned}$$

where $\alpha_1, \alpha_2, \alpha_g > 0$ and $\mu \in \mathbb{R}$ are tuning constants.

Proposition

The solution (ϕ_1, ϕ_2) of the system (4), (12) converges exponentially to the steady state $\phi \equiv 0$ in the X -norm, if

$$e_p > |k_1 k_2| \quad \text{and} \quad |c| < c_{\text{crit}, \text{Lyap}}$$

where, $c_{\text{crit}, \text{Lyap}} > 0$ denotes the critical bound on the coupling coefficient, c , and is given by

$$c_{\text{crit}, \text{Lyap}} = \begin{cases} \sqrt{\frac{|k_1|}{|k_2| e_p}} \lambda^2 \ln\left(\frac{e_p}{k_1 k_2}\right), & \text{if } e_p \leq |k_1 k_2| e^2 \\ \frac{2\lambda^2}{e^1 |k_2|}, & \text{if } e_p > |k_1 k_2| e^2 \end{cases}$$

NUMERICAL RESULTS

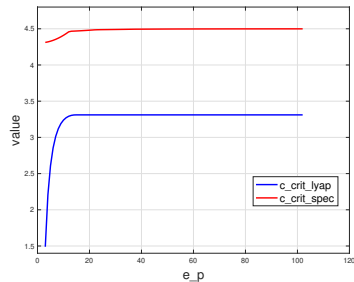
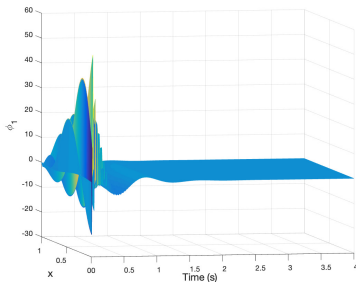
Numerical comparison of the critical parameters $c_{crit,Lyap}$ and $c_{crit,spec}$

$$\begin{cases} \partial_t \phi - \begin{pmatrix} \lambda & c \\ 0 & \lambda \end{pmatrix} \partial_x \phi = 0, & t \in (0, +\infty), x \in (0, 1), \\ \phi_1(t, 1) = k_1 \phi_2(t, 0), \quad \frac{d\phi_2}{dt}(t, 1) = k_2 \phi_1(t, 0) - e_p \phi_2(t, 1). \end{cases} \quad (28)$$

with the parameter values $\lambda = 3$, $k_1 = 1$, $k_2 = 2$, and $e_p = 4$, subject to:

$$\phi_1^0(x) = \sin(2\pi x), \quad \phi_2^0(x) = \cos(2\pi x) - 1, \quad \forall x \in [0, 1].$$

w.r.t these parameters we find out numerically that $c_{crit,spec} = 4.5$, $c_{crit,Lyap} = 2.2$.



(left) State ϕ_1 over the (t, x) -plane (right) Evolution of the critical parameter values as functions of the damping coefficient e_p

The general case of balance laws with nonuniform coefficients

We now consider the case where:

$$\lambda_1(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{2} \\ 2 - \frac{7}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 2, & \text{if } 0 \leq x < \frac{1}{2} \\ 2 + \frac{7}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$c(x) = \frac{1}{10}(x+1)$, $k_1 = 1$, $k_2 = 2$ and $e_p = 5$.

$B_{11}(x) = B_{12}(x) = B_{22}(x) = \frac{x}{200}$, $B_{21}(x) = \frac{\lambda_2(x) - \lambda_1(x)}{200}$. The initial conditions are taken identical to those in the previous section.

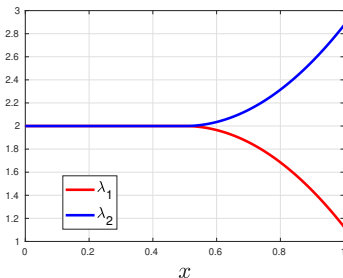
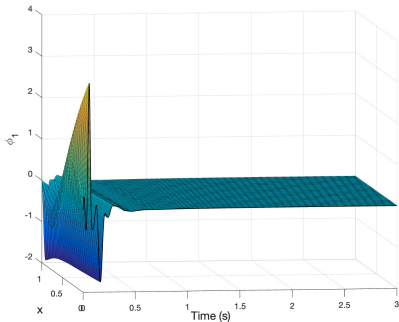
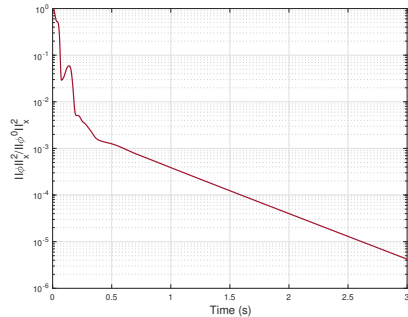


Fig. Characteristic velocities on the spatial domain.



(left) State ϕ_1 over the (t, x) -plane
semi-logarithmic scale



(right) Time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$ in
semi-logarithmic scale

Remarks

Both states are exponentially stabilized to zero as time increases.

⚠ However, with reference to Theorem 3, verifying the feasibility of the matrix inequalities for larger source terms becomes significantly more challenging.

CONCLUSIONS

Study of a class of Non-Strictly Hyperbolic problems:

- Tackling the *new* difficulties related to the non-diagonal structure
- Establishing sufficient stability conditions for this class of systems
- Complete Lyapunov and spectral analysis
- Numerical validation and comparison with theoretical results

⚠ Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory, Advection of tracers in pipes etc.

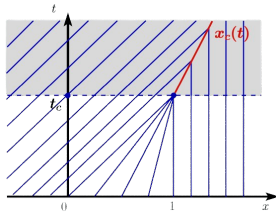
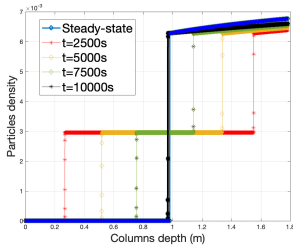
Paper submissions

-  M.O. AMIRAT, V. ANDRIEU, J. AURIOL, M. BAJODEK and C. VALENTIN, *Boundary feedback control of a class of 2×2 weakly hyperbolic systems*, 5th IFAC Workshop on Control of Systems Governed by PDE (CPDE 2025), Beijing, China.
-  M.O. AMIRAT, M. BAJODEK, V. ANDRIEU, J. AURIOL and C. VALENTIN, *Insights into the stability analysis of 2×2 linear weakly hyperbolic systems*. Submitted to *Automatica* (**Status: Under Review**)

PERSPECTIVES: STABILIZATION OF A SHOCK STEADY-STATE

Back to the Clarifier Model

We now aim at **bridging** the previous study w.r.t stabilizing the sludge blanket in the clarifier (that is a *shock steady-state*)



Moving shock and steady-state (top),
Characteristic lines (bottom).

Thus, we are interested in solutions $(\varepsilon_s(t, z), f_s(t, z))^T$ that may have a **jump discontinuity** at some spatial point $z_s(t) \in (0, z_b)$ and are classical elsewhere.

⇒ From the **Rankine-Hugoniot relation** applied to the Clarifier model, two quantities are conserved through the jump. We thus have the following condition at the jump $z_s(t)$:

$$[\varepsilon_s] \dot{z}_s(t) = [f_s]$$

$$[f_s] \dot{z}_s(t) = \left[\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right]$$

Where $\dot{z}_s$ denotes the speed of the jump, and

$$[w] = w(z_s^+(t)) - w(z_s^-(t))$$

Stabilization goal

We want to stabilize ε_s^* , f_s^* and **the shock location** z_s^* .

Ongoing work

First, how to avoid time varying domains $[0, z_s(t)]$ and $[z_s(t), z_b]$?

⇒ Based on the work of G. BASTIN, J.-M. CORON, A. HAYAT and P. SHANG on the stabilizatn of *Hydraulic Jumps*, we perform a transformation of the state coordinate z allowing to define new state variables on the **fixed domain** $[0, z_s^*]$:

$$\varepsilon_{s,1}(t, z) = \varepsilon_s(t, z \frac{z_s}{z_s^*})$$

$$f_{s,1}(t, z) = f_s(t, z \frac{z_s}{z_s^*})$$

$$\varepsilon_{s,2}(t, z) = \varepsilon_s(t, z_b + z \frac{z_s - z_b}{z_s^*})$$

$$f_{s,2}(t, z) = f_s(t, z_b + z \frac{z_s - z_b}{z_s^*})$$

⇒ **Which yields an equivalent 4×4 shock-free system.**

Now consider deviations of the states w.r.t steady-states:

$$F_{s,i} = f_{s,i} - f_s^*, \quad E_{s,i} = \varepsilon_{s,i} - s_i^*(z) \text{ for } i \in 1, 2, \quad Z_s = z_s - z_s^*.$$

Performing Linearization and introducing the following Riemann coordinates:

$$\phi = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ (\frac{f_s^*}{s_1^*})^2 & -\frac{f_s^*}{s_1^*} \end{pmatrix}^{-1} & 0 \\ 0 & \begin{pmatrix} 1 & 1 \\ -\lambda_3 & \lambda_4 \end{pmatrix}^{-1} \end{pmatrix} \begin{pmatrix} E_{s,1} \\ F_{s,1} \\ E_{s,2} \\ F_{s,2} \end{pmatrix} \quad (29)$$

Then the system can be rewritten as a linear weakly hyperbolic system coupled with an ODE:

$$\partial_t \phi - \tilde{A}(z_s^*) \partial_z \phi = \tilde{B}(z) \phi + \tilde{C}(z) \mathbf{Z}_s(t), \quad (30)$$

$$\dot{\mathbf{Z}}_s(t) = \frac{-\lambda_3 \phi_3(t, z_s^*) + \lambda_4 \phi_4(t, z_s^*) - \lambda^2 \phi_1(t, z_s^*) + \lambda \phi_2(t, z_s^*)}{s_2^*(z_s^*) - s_1^*(z_s^*)}, \quad (31)$$

along with jump condition at $z = z_s^*$

$$\lambda^2 \phi_2(t, z_s^*) + (-\lambda^2 + \frac{\sigma'_e(s_2^*)}{\rho_s})(\phi_3(t, z_s^*) + \phi_4(t, z_s^*)) - 2\lambda(\lambda^2 \phi_1(t, z_s^*) - \lambda \phi_2(t, z_s^*)) + 2\lambda(-\lambda_3 \phi_3(t, z_s^*) + \lambda_4 \phi_4(t, z_s^*)) = 0.$$

where

$$\tilde{A}(z_s^*) = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda_3 \frac{z_s^*}{z_b - z_s^*} & 0 \\ 0 & 0 & 0 & \lambda_4 \frac{z_s^*}{z_b - z_s^*} \end{pmatrix}, \text{ and } \tilde{B}, \tilde{C} \text{ are matrices of class } C^1.$$

Upper block falls within the scope of the previously studied system (CPDE2025 + Automatica paper)

Finally, we consider the following set of boundary conditions:

$$\left\{ \begin{array}{l} \phi_1(t, 0) = k_1 \phi_2(t, z_s^*) \\ \frac{d}{dt} \phi_2(t, 0) = k_2 \phi_1(t, z_s^*) - e_p \phi_2(t, 0) \\ \phi_3(t, 0) = \frac{\lambda_4}{\lambda_3} \phi_4(t, 0) - \frac{\lambda^2}{\lambda_3} \phi_1(t, z_s^*) + \frac{\lambda}{\lambda_3} \phi_2(t, z_s^*) \\ \phi_4(t, z_s^*) = \frac{2\lambda^3 \phi_1(t, z_s^*) - 3\lambda^2 \phi_2(t, z_s^*) - \left[-\lambda^2 + \frac{\sigma'_e(s_2^*)}{\rho_s} - 2\lambda\lambda_3 \right] \phi_3(t, z_s^*)}{-\lambda^2 + \frac{\sigma'_e(s_2^*)}{\rho_s} + 2\lambda\lambda_4} \end{array} \right. \quad (32)$$

Along with the initial conditions

$$\begin{aligned} \phi(0, x) &= \phi_0(x) = (\phi_{10}(x), \phi_{20}(x), \phi_{30}(x), \phi_{40}(x))^T, \\ Z_s(0) &= Z_{s,0} \end{aligned} \quad (33)$$

Thus, system (30),(31),(32),(33) is a PDE-ODE coupling.

In order to study the exponential stability of the steady-state $(\mathbf{0}_{\mathbb{R}^4}, 0)$ we adopt a *forwarding approach* (**Ongoing work...**):

- 1) Ignore the ODE coupling (i.e. $\tilde{C} = 0$) and establish stability conditions for the PDE system.
- 2) By robustness and continuity arguments we add the ODE effects.

Thank you for your attention
Questions ?